Module 3: Customer Problem Statements & System Requirements
INFO C450 - System Design
Spring 2026
Titus Duncan II

Collaborative Media Management System, Titus Duncan

1.1 Customer Problem Statement:

"As a smaller-market creative, I don't have a dedicated tool to manage projects and clients. I'm just trying to figure it out, so I use what's common and easy; things like Google Drive folders, texts, emails, and whatever link-sharing works in the moment. But when my work is spread across different tools, organization falls apart. Files get buried, overwritten, corrupted, or lost. I send one version, then later realize the client reviewed a different one. Feedback ends up scattered across messages, and approvals are unclear. I waste time hunting for the latest deliverable instead of creating, and it makes me look less professional than I actually am.

What I expect from a software system is simple: keep everything organized on one dedicated platform. I want one devoted workspace per project where I can upload and label my assets, track revisions, and know exactly what version is current. I want client feedback attached to the correct file, and I want a clean presentation page I can share so clients can review deliverables in order and clearly approve or request changes. If a system can make it simple to organize—so nothing gets lost and everyone stays on the same page—then collaboration becomes smoother, revisions become faster, and delivering final work becomes much more reliable."

1.2 Glossary Terms and Visual Representation:

Term	Definition
Project	A dedicated space that keeps all work for one job or collaboration together in one place.
Collaborator	A person invited into a project to contribute work, revisions, or feedback.
Client	The person the work is for. In some projects, the client may also be a collaborator.
Asset	A file saved in a project (image, video, audio, or document).
Version	A saved revision of an asset (v1, v2, v3…) so earlier work is not overwritten or lost.
Current Version	The version that should be treated as the main one for review or delivery at that time.
Status	The stage a version is in: Draft, In Review, Changes Requested, or Approved/Final.
Comment	Feedback attached to a specific version, so notes stay connected to the correct file.
Presentation Page	A client-facing page that displays selected deliverables in an organized order for review.
Share Link	A controlled access link to a presentation page (view-only or comment/approve, optional expiration).

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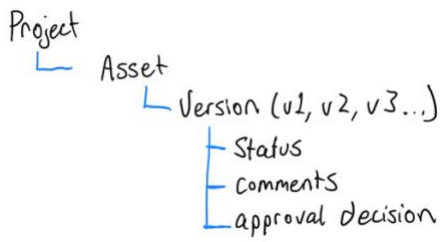
1.2.1 Information Structure Examples:

Project Structure



Project in Review Flow

Client Presentation Flow

1.3 System Requirements (Functional & Non-Functional):

Key Term	Meaning
Req-#	Functional Requirement
NFR-#	Non-functional Requirement
High	the core promise of "organization is key" (projects, assets, versions, current status, feedback).
Medium	review + presentation layer (important, but the system can still function as an organized collaboration vault without all client-facing features complete).
Low	search/filtering (nice-to-have early; becomes critical later as volume grows).

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System Requirements

No.	**Priority Weight**	**Description**
REQ-1	High	Users need to create an account to sign in and have access.
REQ-2	High	The system must support **projects** so work stays organized by job or collaboration.
REQ-3	High	A project owner must be able to invite and manage **collaborators** for each project.
REQ-4	High	The system must support uploading **assets** into a project.
REQ-5	High	The system must support **versioning** so revisions don't overwrite or replace older work.
REQ-6	High	Each version needs a **status** (Draft, In Review, Changes Requested, Approved/Final) and a clear **current version**.
REQ-7	High	**Comments** must attach to a specific version so feedback stays tied to the right file.
REQ-8	Medium	**Approval decisions** must be recorded on a version (approve or request changes) to prevent confusion.
REQ-9	Medium	Assets should include organizing details (title, description, tags, type) so projects don't turn into messy file dumps.
REQ-10	Medium	The system should support a **presentation page** that shows selected deliverables in a clear order.
REQ-11	Medium	A **share link** should provide controlled access to a presentation page (view-only or comment/approve).
REQ-12	Low	Search and filters (keyword/tag/type/status) are helpful for speed as projects grow.
NFR-1 (Usability)	High	The system must feel simple and organized for first-time users, with clear labels and an obvious flow from project → asset → version → feedback/approval.
NFR-2 (Reliability)	High	The system must protect work from being lost or overwritten by saving assets, versions, comments, and approvals consistently and keeping version history intact.
NFR-3 (Functionality)	High	Access control must be dependable so only invited collaborators and shared clients can view a project or presentation page.
NFR-4 (Performance)	Medium	Common actions (opening a project, viewing an asset/version, loading a presentation page) should feel responsive for typical class-scale use.
NFR-5 (Usability)	Medium	The system should be easy to navigate on a laptop screen without training, including readable layouts for reviewing deliverables and leaving comments.
NFR-6 (Supportability)	Medium	The project must be easy to maintain and grade: clear folder structure, readable naming, and consistent terms across the UI, database, and documentation.
NFR-7 (Supportability)	Medium	The system should be easy to set up on a new machine using documented steps (install, configure, run, test).
NFR-8 (Performance)	Low	The system should handle moderate growth (more projects/assets) without needing major redesign, as long as users follow basic organization (tags, projects, versions).

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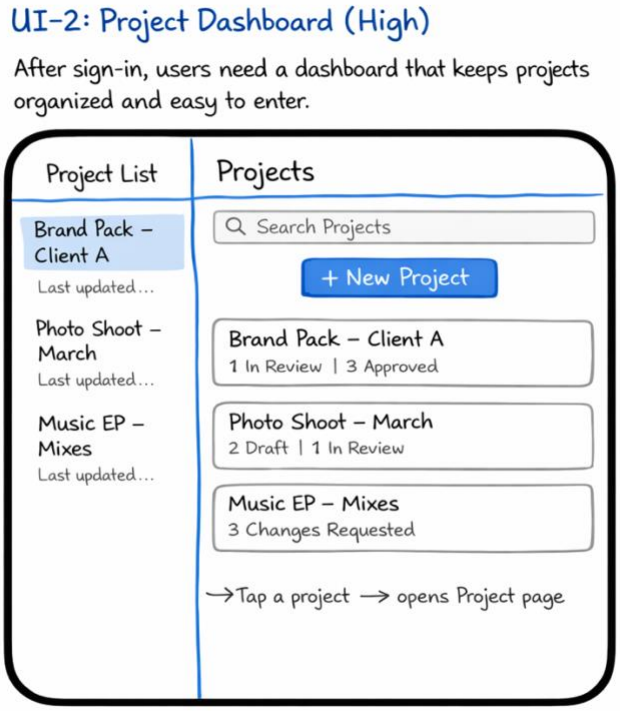
1.4 User Interface Requirements:

Disclaimer: I hand drew these images on my iPad, then enhanced them using AI.



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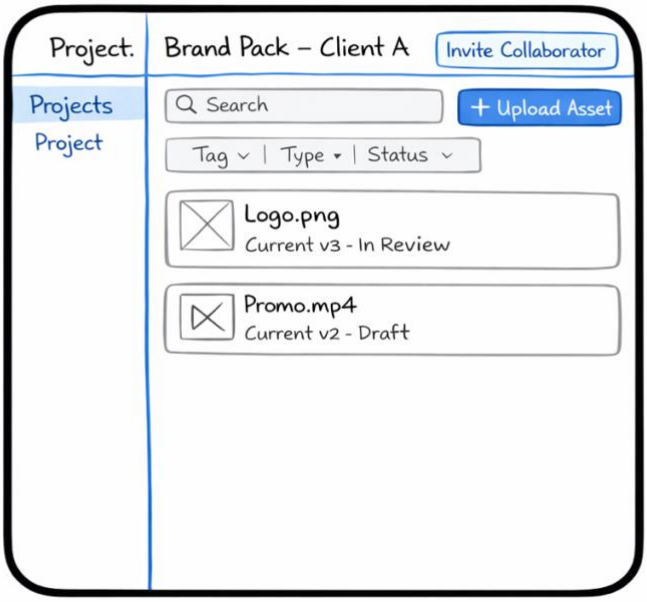
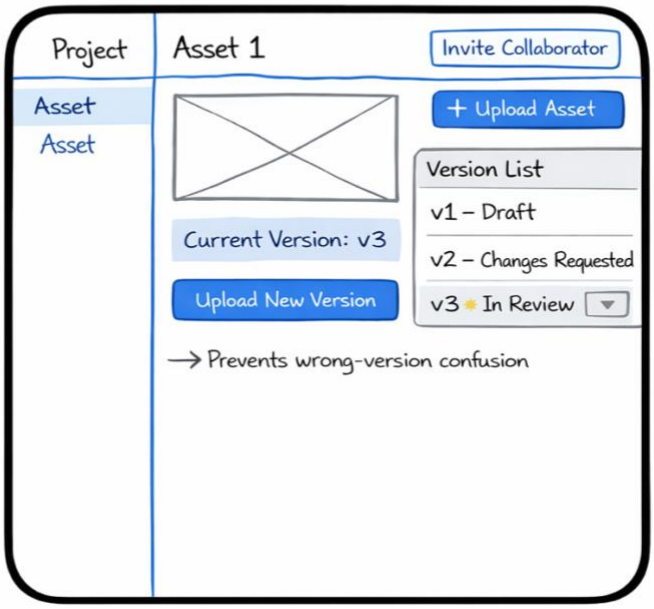
UI-3: Project Library (High)

Inside a project, users need to upload assets and browse everything in one place.



UI-4: Asset + Versions (High)

Users need a clear asset page that makes the current version and past versions obvious to prevent confusion.



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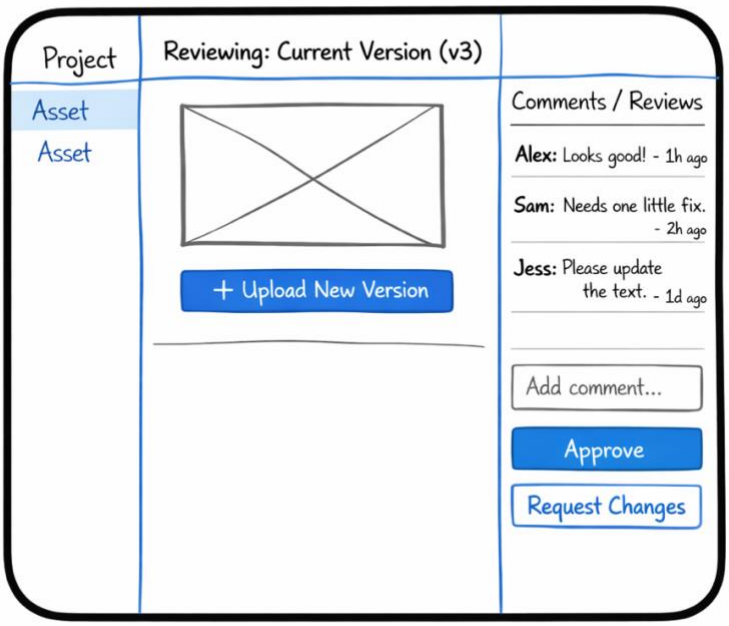
UI-5: Comments + Approval (High)

Users need to review an asset version and make approval decisions.



UI-6: Presentation Page (Medium)

Users need to share a clean page for client review.

1.5 Plan of Work Summary:

So far, I've completed the following:

- Project Topic
- Customer problem statement
- Glossary of terms
- Functional and Non-functional requirements
- User Interface Requirements
- Basic Sketches

I am currently in the process of:

- Setting up development environment
- Creating Django project structure
- Starting database design

My next focus is:

- Finish environment setup and assure that everything runs local properly
- Build the first working flow as mentioned in section 1.2.1
- Begin database models and connect the UI flow to database actions